

FORMAL LANGUAGES AND COMPILERS

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Exam of Tue 21 February 2017 - Part Theory

LAST + FIRST NAME:

(capital letters please)

MATRICOLA:

(or PERSON CODE)

SIGNATURE:

INSTRUCTIONS - READ CAREFULLY:

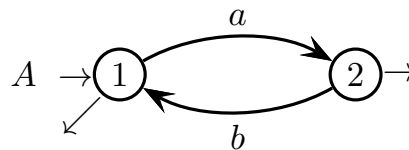
- The exam is in written form and consists of two parts:
 1. Theory (80%): Syntax and Semantics of Languages
 - regular expressions and finite automata
 - free grammars and pushdown automata
 - syntax analysis and parsing methodologies
 - language translation and semantic analysis
 2. Lab (20%): Compiler Design by Flex and Bison
- To pass the exam, the candidate must succeed in both parts (theory and lab), in one call or more calls separately, but within one year (12 months) between the two parts.
- To pass part theory, the candidate must answer the mandatory (not optional) questions; notice that the full grade is achieved by answering the optional questions.
- The exam is open book: textbooks and personal notes are permitted.
- Please write in the free space left and if necessary continue on the back side of the sheet; do not attach new sheets and do not replace the existing ones.
- Time: part lab 60m - part theory 2h.15m

1 Regular Expressions and Finite Automata 20%

1. Consider the following regular expression e , over the two-letter alphabet $\Sigma = \{ a, b \}$:

$$e = (a b \mid a)^+ (b a)^*$$

Consider also the finite-state automaton A below, over the same alphabet Σ , with initial state 1 and final states 1, 2:



Answer the following questions:

- (a) Find the shortest ambiguous string of the regular language $L(e)$ and suitably justify your answer. Provide evidence that the string is ambiguous and that it is the shortest ambiguous one.
 - (b) By using the Berry-Sethi method, design a finite-state deterministic automaton B that accepts the language $L(e)$, and if necessary minimize the automaton B by applying the suitable minimization algorithm.
 - (c) By using a systematic method, design a finite-state automaton C that accepts the intersection language $L(e) \cap L(A)$.
 - (d) By using the node elimination method (Brzozowsky) applied to automaton A , find an equivalent regular expression r .
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2 Free Grammars and Pushdown Automata 20%

1. Consider a two-letter alphabet $\Sigma = \{ a, b \}$ and the classical free language L of the palindromes without centre, below:

$$L = \{ x x^R \mid x \in \Sigma^* \}$$

Answer the following questions:

- (a) Write a *BNF* grammar G_1 (no matter if ambiguous) that generates the language L_1 of the (possibly empty) prefixes of language L , namely:

$$L_1 = \{ y \in \Sigma^* \mid \exists z \in \Sigma^* \text{ such that } yz \in L \}$$

- (b) Write a *BNF* grammar G_2 (no matter if ambiguous) that generates the language L_2 of the (possibly empty) substrings of language L , namely:

$$L_2 = \{ w \in \Sigma^* \mid \exists u, v \in \Sigma^* \text{ such that } uwv \in L \}$$

- (c) (optional) Determine if grammars G_1 and G_2 are ambiguous and justify your answer: if a grammar is ambiguous then provide a string with two or more syntax trees, else argue that it does not generate any ambiguous string.
-

2. Consider a description language that models the organizational chart of a generic institution, which is structured according to the following specifications:

- At the top level, the organization has a name (a non-empty alphanumerical string), a president, a secretary and a staff (a non-empty group of participating persons); all are mandatory.
- The organization must also include a non-empty list of committees. A committee may include one or more subcommittees, and a subcommittee may further include one or more subcommittees, at any depth level.
- A committee has a name (a non-empty alphanumerical string), a chairperson, a secretary, a staff (a non-empty group of participating persons), an establishment date and a dismissal one; all are mandatory, except for the dismissal date.
- A subcommittee has the same structure as a committee, except that the secretary and the establishment date are optional, and that a dismissal date may be specified provided that an establishment date is specified as well.
- The order of the various elements listed above in the organization, in the committees and in the subcommittees, is fixed and is shown in the example below.
- A person is described by his first name and his last name (both names are alphabetical strings), and by his pid (personal identifier), which is a string of exactly six decimal digits. A date is composed of a day/month/year in the customary format xx/xx/xxxx, where x is a decimal digit.

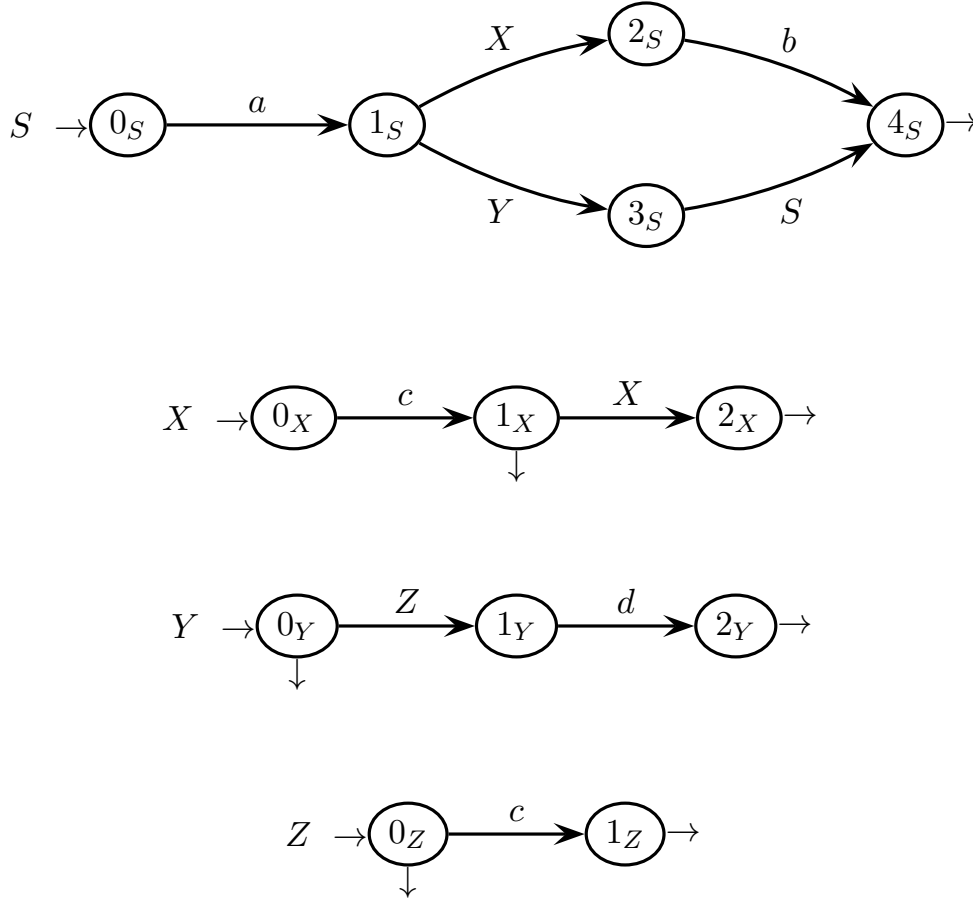
Further detailed information about the lexical and syntactic structure of the description can be inferred from the example below:

```
organization CompSciConference2017
  president firstname Lilli lastname Verdi pid 839257;
  secretary firstname Paolo lastname Bianchi pid 325476;
  staff
    firstname ... lastname ... pid ...;
    firstname ... lastname ... pid ...;
  end staff
  committee communication
    chairperson firstname Mario lastname Sarti pid 937294;
    secretary firstname Lisa lastname Volo pid 739473;
    staff ... end staff
    establishment 23/07/2015;
    subcommittee invitations
      chairperson firstname Sandra lastname Piano pid 938402;
      staff ... end staff
      establishment 14/08/2015;
      dismissal 21/01/2017;
    end subcommittee invitations
  end committee communication
  ...
end organization
```

Write a grammar, *EBNF* and unambiguous, that models the sketched language.

3 Syntax Analysis and Parsing Methodologies 20%

1. Consider the following grammar G , represented as a machine net over the four-letter terminal alphabet $\Sigma = \{ a, b, c, d \}$ and the four-letter nonterminal alphabet $V = \{ S, X, Y, Z \}$ (axiom S):

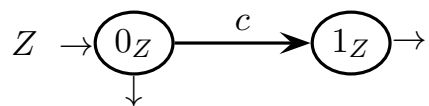
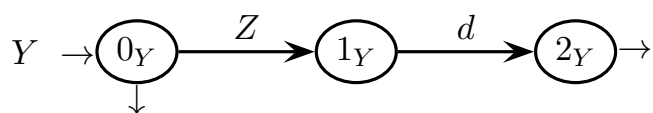
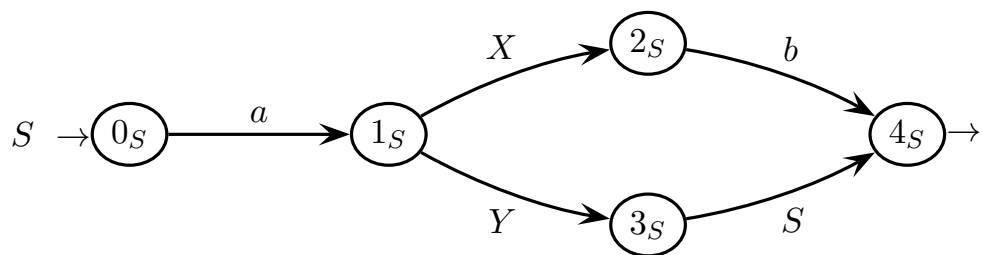
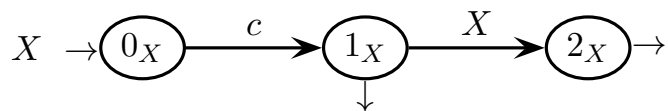


Answer the following questions:

- (a) Draw the complete pilot of grammar G , say if grammar G is of type $ELR(1)$, and shortly justify your answer. If grammar G is not $ELR(1)$ then highlight all the conflicts in the pilot.
- (b) Write the necessary guide sets on the arcs of the machine net, show that grammar G is not of type $ELL(1)$, based on the guide sets, and shortly justify your answer (for the guide sets please use the figure prepared on the next page).
- (c) Say if grammar G is of type $ELL(k)$ for some $k > 1$, and provide an adequate justification to you answer.
- (d) (optional) Identify the four languages $L(Z)$, $L(Y)$, $L(X)$ and $L(S)$ by resorting, for representing each of them, to a formalism as simple as possible. Determine if there is a grammar G' equivalent to grammar G that is of type $ELL(1)$, and adequately justify your answer.

please here draw the pilot graph

please here draw the call arcs and write the guide sets with $k = 1$



4 Language Translation and Semantic Analysis 20%

1. The grammar G below (axiom S) generates bit strings that represent positive natural numbers according to the usual positional binary encoding:

$$G \left\{ \begin{array}{l} S \rightarrow S0 \mid S1 \mid N1 \\ N \rightarrow N0 \mid N1 \mid \varepsilon \end{array} \right.$$

For instance, string 010100 represents in binary the (decimal) number 20, string 1010 represents number 10 and string 011 number 3.

Please answer the following questions:

- (a) Define a translation scheme that concatenates to the source string a prefix consisting of a number of symbols ‘ e ’ equal to the exponent of the largest power of 2 of which the encoded number is a multiple.

For instance, the string 010100 encodes number 20, which is a multiple of 4 (but not a multiple of 8), thus it is translated into string $ee010100$. Other examples are $1010 \mapsto e1010$ and $011 \mapsto 011$. If necessary, modify the source grammar as little as possible.

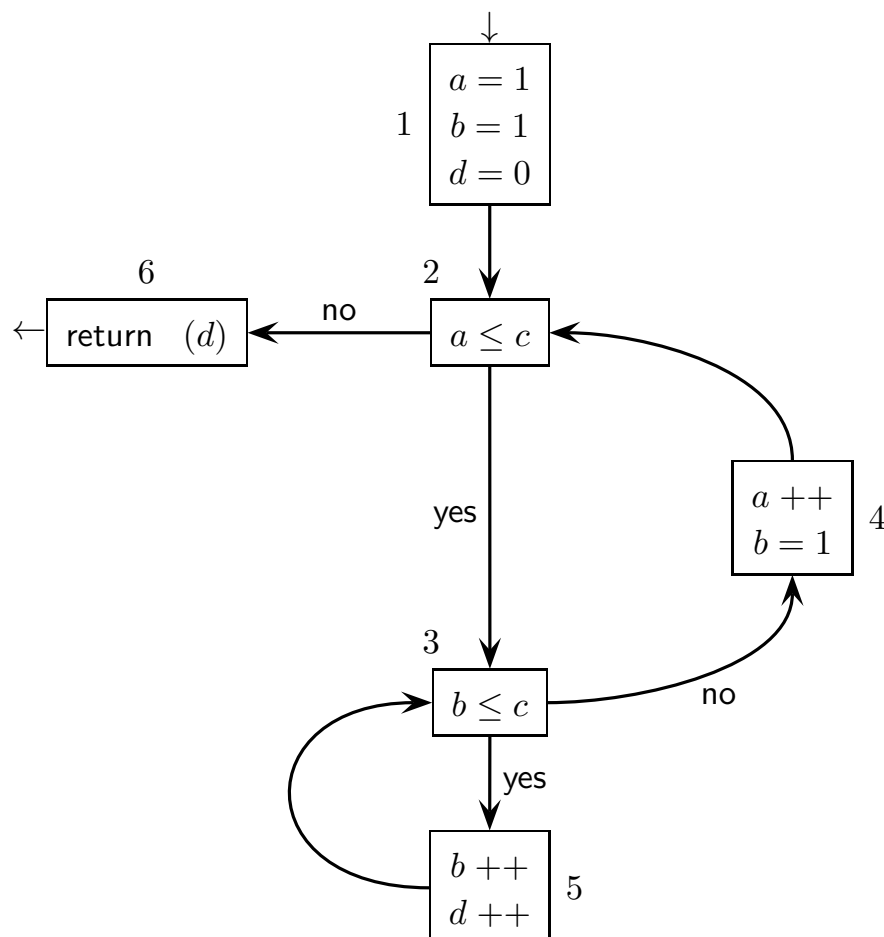
Can this translation be modeled with a rational (regular) translation scheme ? In the negative case, please justify your answer; in the positive case, provide such a regular translation scheme.

- (b) Define a translation scheme that cancels all the leading zeroes from the source string. For instance: $010100 \mapsto 10100$, $1010 \mapsto 1010$ and $011 \mapsto 11$. If necessary, modify the source grammar as little as possible.

Can this translation be modeled with a rational (regular) translation scheme ? In the negative case, please justify your answer; in the positive case, provide such a regular translation scheme.

- (c) Is the translation of the previous point (b) surjective ? Is it injective ? To answer these questions, consider the source and destination languages as domain and image of the translation function, respectively. Provide an adequate justification to your answer.

2. Consider the following Control Flow Graph (CFG) of a program, with some nodes that contain multiple assignments (input node 1 and output node 6):



This program has three variables a , b and d , and an input parameter c . All of them are integers ≥ 0 .

Answer the following questions:

- Find a regular expression R , over the node name alphabet, that describes the execution traces of the program, ignoring any semantic restriction.
- Directly identify the variables live at the node inputs of the CFG and write them on the CFG (use the figure prepared on the next page).
- Write the system of flow equations for the live variables of the CFG, and iteratively solve the system (use the tables prepared on the next pages). Verify that the obtained solution coincides with that found at point (b).
- (optional) Determine the value d returned by the program as a function of the input parameter c . Based on this function, refine the regular expression R found at point (a) into a language expression over the node name alphabet that exactly defines the program execution traces. Is the language of the execution traces still regular? Adequately justify your answer.

please write here the regular expression R

please write here the variables live at the node inputs

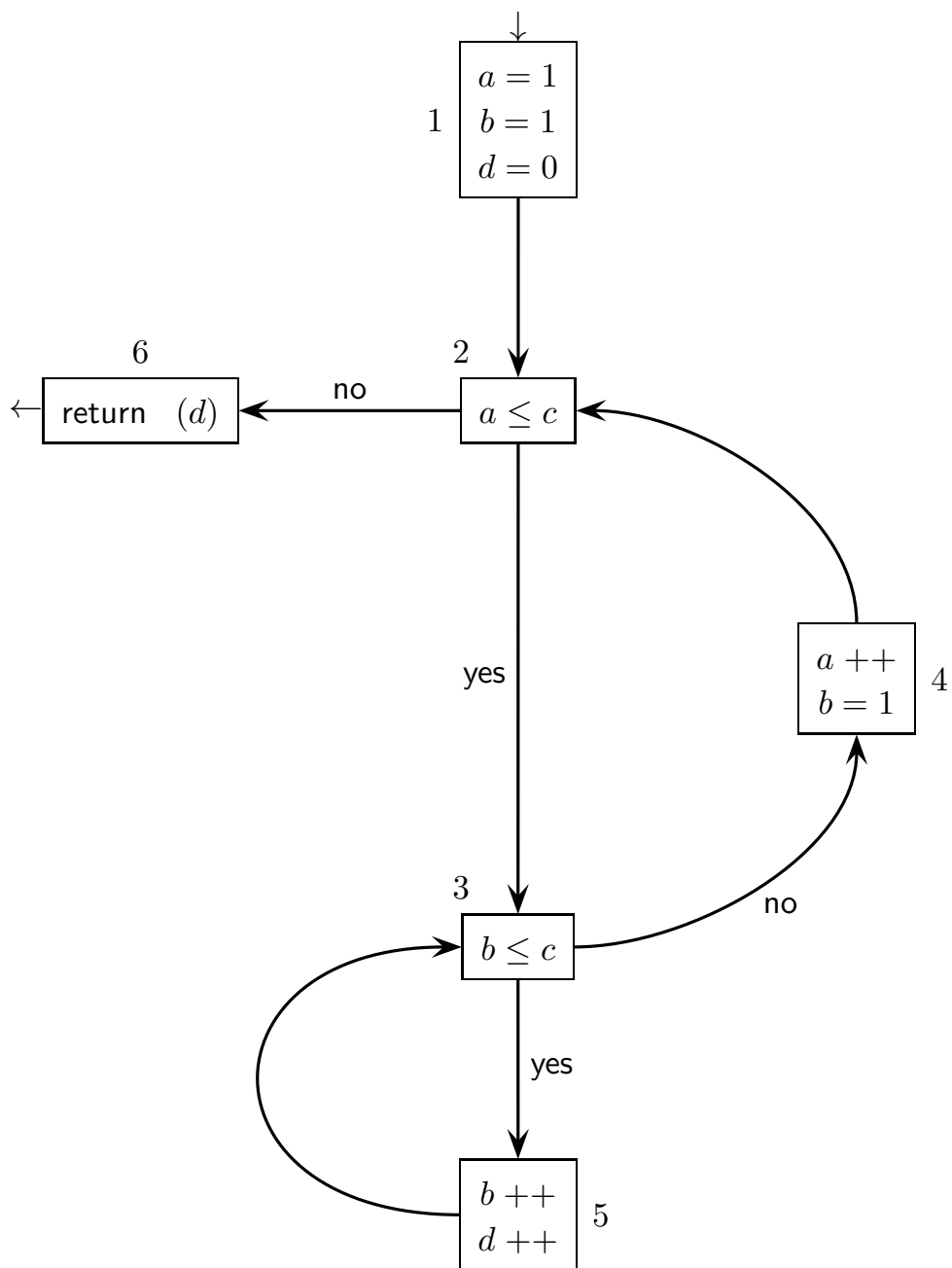


table of definitions and usages at the nodes

<i>node</i>	<i>defined</i>	<i>node</i>	<i>used</i>
1		1	
2		2	
3		3	
4		4	
5		5	
6		6	

system of data-flow equations

<i>node</i>	<i>in equations</i>	<i>out equations</i>
1		
2		
3		
4		
5		
6		

iterative solution table of the system of data-flow equations
 (the number of columns is not significant and is more than sufficient to solve the system)

	<i>initialization</i>		1		2		3		4		5	
#	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>	<i>out</i>	<i>in</i>
1												
2												
3												
4												
5												
6												

